

Assignment - 4

UCB Algorithm

Given

Machines = A, B, C

Number of plays: 10

Reward Pattern

<u>Machine</u>	<u>Rewards</u>
A	3, 4, 2, 5
B	6, 7, 5, 8
C	4, 3, 5, 4

$$\text{UCB: } \text{UCB}(a) = Q(a) + \sqrt{\frac{2 \ln(t)}{N(a)}}$$

Incremental Q-value:

$$Q_{\text{new}} = Q_{\text{old}} + \frac{R - Q_{\text{old}}}{N(a)}$$

Play 1:

Initially:

$$N(A) = N(B) = N(C) = 0$$

$$\therefore \text{UCB}(A) = \text{UCB}(B) = \text{UCB}(C) = \infty$$

Machine A is selected

Reward from A:

$$R = 3$$

$$N(A) = 1$$

$$Q(A) = 0 + \frac{3 - 0}{1}$$

$$Q(A) = 3$$

After play 1:

Machine

	<u>N</u>	<u>R</u>	<u>Q</u>
A	1	3	3
B	0	0	0
C	0	0	0

play 2:

for machine A:

$$UCB(A) = 3 + \sqrt{\frac{2 \ln(2)}{1}}$$

$$\therefore \ln(2) = 0.6931$$

$$UCB(A) = 3 + \sqrt{2(0.6931)}$$

$$= 3 + \sqrt{1.3862}$$

$$UCB(A) = 4.1774$$

Machines B and C have not been selected yet.

$$UCB(B) = \infty$$

$$UCB(C) = \infty$$

so machine B is selected

Reward from B:

$$R = 6$$

$$N(B) = 1$$

$$Q(B) = 0 + \frac{6-0}{1}$$

$$\boxed{QCB} = 6$$

After play 2:

<u>Machine</u>	<u>N</u>	<u>R</u>	<u>Q</u>
A	1	3	3
B	1	6	6
C	0	0	0

Total rewards after 2 plays

$$\text{Total} = 3 + 6$$

$$\boxed{\text{Total} = 9}$$

Total rewards after 10 plays

<u>play</u>	<u>Machine</u>	<u>Reward</u>
1	A	3
2	B	6
3	C	0
4	B	6
5	B	5
6	B	8
7	B	6
8	B	7
9	B	5
10	B	8

$$\text{Total Reward} = 3 + 6 + 4 + 7 + 5 + 8 + 6 + 7 + 5 + 8 +$$

$$= 59$$

$$\boxed{\text{Total reward} = 59}$$

Final Result:

<u>Machine</u>	<u>No. of plays (N)</u>	<u>Total Rewards (R)</u>	<u>Q-value</u>
A	1	3	3.00
B	8	52	6.50
C	1	4	4.00
<u>Total:</u>	<u>10</u>	<u>59</u>	

Conclusion:

- 1) Total reward received = 59
- 2) Best machine = B
- 3) Highest Q-value = 6.50